

ULB creditcard.csv

284,807 tx · 492 fraud (0.17%)
28 PCA features + Time + Amount

PaySim

6,362,620 tx · 8,213 fraud (0.13%)
raw financial features + type (one-hot)

Chronological 60/20/20 split (ULB: sort by Time · PaySim: sort by step)

train fit the detectors **validation** fit the decision rule **test** scored once, never used to fit or to threshold

Preprocessing (ULB)

RobustScaler(Amount, Time, Hour)
hour-of-day feature
fitted on the training split only

Preprocessing (PaySim)

RobustScaler + global StandardScaler
balance residuals $\delta_{\text{orig}}, \delta_{\text{dest}}$
fitted on the training split only

Three fitting regimes, all sharing the same split and the same decision rule

U label-free: fit on the full train, ρ from an a-priori grid **N** normal-only: fit on legitimate train rows
O oracle: as U but $\rho = \text{true train prevalence}$ — a reference, not deployable

Five detectors, identical pipeline on both datasets

Isolation Forest | LOF (novelty) | One-Class SVM | K-Means
DBSCAN, scored out of sample as the distance to the nearest training core point
scores only — no detector emits a decision

Calibration on the validation window

threshold $\tau_\rho = Q_{1-\rho}$ of the *validation* scores · score $\rightarrow$ validation ECDF
applied unchanged to the test stream: a decision depends on one transaction, never on the batch

Ensemble

mean of the validation ECDFs
(five detectors, DBSCAN included)

Hybrid bridge

scores $\rightarrow$ logistic regression
(consumes validation labels)

Imbalance-aware evaluation

Precision · Recall · F_1 · ROC-AUC · **PR-AUC (primary)** · alert volume · per-transaction latency
+ rolling origins + prevalence-matched random arm + block bootstrap
+ ablation of the clock inputs and of the PaySim balance residuals
+ supervised reference (PR-AUC 0.78–0.81 on ULB)